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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

AI-ASSISTED HEALTH-RELATED MISINFORMATION VERIFICATION SYSTEM

A PROJECT REPORT

Submitted in partial fulfillment of the requirements for the award of the degree of

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Submitted by:

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ACADEMIC YEAR: [2025 – 2026]

BONAFIDE CERTIFICATE

This is to certify that the project report entitled 'AI-ASSISTED HEALTH-RELATED MISINFORMATION VERIFICATION SYSTEM' is the bonafide work of [STUDENT NAME] (Register Number: [STUDENT REGISTER NUMBER]), who carried out the project work under my supervision in partial fulfillment of the requirements for the award of the degree of Bachelor of Engineering / Bachelor of Technology in Computer Science and Engineering at [COLLEGE NAME], affiliated with [UNIVERSITY NAME], during the academic year [2025–2026].

To the best of our knowledge and belief, this project report embodies original work completed by the candidate and has not formed the basis for the award of any other Degree, Diploma, Associateship, Fellowship, or similar title in this or any other educational institution.

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CANDIDATE'S DECLARATION

I hereby declare that the project work presented in this report entitled 'AI-ASSISTED HEALTH-RELATED MISINFORMATION VERIFICATION SYSTEM' submitted to the Department of Computer Science and Engineering, [COLLEGE NAME], is an authentic record of my own work carried out under the guidance of [PROJECT GUIDE NAME], [DESIGNATION], Department of Computer Science and Engineering.

I further declare that this report has not been submitted previously in part or in full to any other University or Institution for the award of any academic degree, diploma, or certificate. All literature, research papers, open-source datasets, and software libraries referenced in this work have been properly acknowledged and cited in accordance with standard academic guidelines.

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ABSTRACT

The proliferation of unverified health assertions across social networks and digital platforms poses severe hazards to public well-being. Misleading claims concerning diseases, unauthorized therapies, and pharmaceutical efficacy often spread rapidly before qualified medical professionals can publish authoritative refutations. While modern generative Large Language Models (LLMs) can produce fluent conversational responses, their tendency to hallucinate unsupported medical statements and their lack of verifiable source citations make them unsuitable as unsupervised truth adjudicators in safety-critical domains.

To address these limitations, this project presents the 'AI-Assisted Health-Related Misinformation Verification System', a deterministic, multi-stage, evidence-grounded verification framework that eliminates unconstrained text generation from the truth determination process. The system retrieves candidate evidence from two complementary authoritative repositories: accredited journalistic fact-checking organizations via the Google Fact Check Tools API and peer-reviewed biomedical literature via the NCBI PubMed E-Utilities API. To ensure precise candidate filtering, the system implements domain-tailored neural re-ranking: a Sentence-BERT (SBERT) bi-encoder (all-MiniLM-L6-v2) for claim reviews and a deep MedCPT Cross-Encoder (ncbi/MedCPT-Cross-Encoder) pre-trained on 255 million PubMed query-article pairs for scientific abstracts.

The Top-10 ranked documents from each source stream undergo directional stance classification using a large pre-trained Natural Language Inference (NLI) model (cross-encoder/nli-deberta-v3-large) under an uncontaminated premise formulation. The resulting stance distributions (entailment, contradiction, and neutral) are synthesized by a deterministic decision engine enforcing a contradiction-precedent safety hierarchy. Evaluated on an expanded benchmark of 60 rigorous test claims (combining curated public health claims and gold SciFact biomedical instances), the system achieved an overall classification accuracy of 83.33%, a macro F1-score of 78.52%, and a weighted F1-score of 83.00%, outperforming the standard SBERT semantic similarity baseline (46.67% accuracy) by +36.66% absolute gain. On the SciFact biomedical retrieval benchmark, MedCPT demonstrated perfect top-rank retrieval with a Mean Reciprocal Rank (MRR) of 1.0000. All findings are rendered within an interactive Streamlit web dashboard featuring full citation traceability to PubMed identifiers (PMIDs) and verified URLs, enabling everyday users to easily check online health advice.

Keywords: Health Misinformation, Natural Language Inference, MedCPT Cross-Encoder, Biomedical Information Retrieval, Sentence-BERT, Decision Engine.

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LIST OF ABBREVIATIONS

Abbreviation / Term	Expanded Form / Definition
AI	Artificial Intelligence
API	Application Programming Interface
BERT	Bidirectional Encoder Representations from Transformers
Bi-Encoder	Two-tower neural architecture encoding sequences independently
CDC	Centers for Disease Control and Prevention
CLS	Classification Token in Transformer architectures
Cross-Encoder	Joint sequence neural architecture encoding pair sequences with cross-attention
DeBERTa	Decoding-enhanced BERT with Disentangled Attention
DFD	Data Flow Diagram
DOI	Digital Object Identifier
E-Utilities	Entrez Programming Utilities API provided by NCBI
F1	Harmonic mean of Precision and Recall (F1-score)
GFC	Google Fact Check Tools API
IFCN	International Fact-Checking Network
IR	Information Retrieval
JSON	JavaScript Object Notation
LLM	Large Language Model
MedCPT	Medical Contrastive Pre-trained Transformers
MEDLINE	Medical Literature Analysis and Retrieval System Online
MRR	Mean Reciprocal Rank
NCBI	National Center for Biotechnology Information
NDCG	Normalized Discounted Cumulative Gain
NLM	National Library of Medicine
NLI	Natural Language Inference
NLP	Natural Language Processing
P@K	Precision at Rank Cut-off K

PMID	PubMed Identifier
PubMed	Free biomedical bibliographic search engine by NLM
R@K	Recall at Rank Cut-off K
REST	Representational State Transfer
SBERT	Sentence-BERT
SEP	Separator Token in Transformer architectures
UI	User Interface
URI	Uniform Resource Identifier
URL	Uniform Resource Locator
WHO	World Health Organization
XML	Extensible Markup Language

CHAPTER 1 – INTRODUCTION

1.1 Introduction

Today, almost everyone turns to the internet when they have questions about their health. Whenever people feel unwell, experience common symptoms, or want advice on diet, fitness, and home remedies, they routinely search online or read posts on social media platforms like WhatsApp, YouTube, and Facebook. Along with helpful advice, however, a large amount of false and misleading health information spreads online every single day. Everyday users frequently encounter viral posts claiming that simple kitchen ingredients can cure serious conditions like asthma or diabetes, or rumors warning people against safe medicines and vaccines. Believing these false claims can be very dangerous. An ordinary person might stop taking the medicines prescribed by their doctor, delay visiting a hospital during an emergency, or try untested home mixtures that could harm their health.

For an everyday person, figuring out whether an online health tip is true or false is very difficult. Most scientific research papers are written using complicated medical language, doctor terminology, and complex statistical figures that only healthcare specialists can easily understand. At the same time, simple internet searches often show conflicting blog posts, sponsored advertisements, and unverified personal opinions, leaving regular people confused about what to trust. Ordinary people simply do not have the time or specialized background knowledge to read through dense clinical studies just to check if a simple health tip they heard from a friend or saw in a family group chat is safe and accurate.

To solve this problem and help the general public, this project introduces the 'AI-Assisted Health-Related Misinformation Verification System'. This system is an easy-to-use tool designed specifically for ordinary people who want a fast, simple, and reliable way to check health claims they see online. When a user enters a question or claim—such as 'Does ginger tea cure asthma?'—the system automatically searches two trusted sources: verified fact-checking records (like Snopes and Health Feedback) and reliable medical research from PubMed. The system uses artificial intelligence to compare the meaning of the claim with the evidence, checks whether scientific findings agree with or disprove the claim, and gives the user a clear and simple result, such as 'Potentially supported', 'Potentially refuted', or 'Insufficient evidence'. By providing straightforward explanations and direct links to trusted sources, the system helps regular people make safer, better-informed health choices.

> **SAFETY PRINCIPLE:** Core Design Principle: The verification system deliberately excludes unconstrained generative text models from the truth adjudication loop. Final verdicts are produced strictly through deterministic aggregation rules applied over calibrated cross-attention inference probabilities and peer-reviewed literature citations.

1.2 Background of the Project

Automated fact-checking has evolved substantially over the past decade. Early verification tools relied heavily on basic keyword filtering and lexical matching against small, static databases of debunked myths. While lexical methods execute quickly, they are fundamentally incapable of understanding linguistic variations, synonyms, or contextual negations. For example, a search for the claim 'Vitamin C prevents COVID-19' against a document containing 'Clinical trials show Vitamin C does not prevent COVID-19' would return a false-positive lexical match because the major keywords overlap, despite the fact that the document directly refutes the assertion.

With the emergence of deep learning and transformer-based language representations such as BERT, Natural Language Processing (NLP) systems gained the ability to capture semantic meaning. Researchers introduced Natural Language Inference (NLI) frameworks, which model the relationship between a premise text and a hypothesis text into three distinct categories: entailment (the premise supports the hypothesis), contradiction (the premise refutes the hypothesis), or neutral (the premise provides neither sufficient support nor refutation).

In the biomedical domain, verifying assertions requires consulting primary scientific evidence. The National Center for Biotechnology Information (NCBI) hosts the PubMed database, which indexes over 36 million citations from MEDLINE, life science journals, and online books. In parallel, global fact-checking organizations (such as Health Feedback, Snopes, PolitiFact, and Full Fact) adhere to the standards of the International Fact-Checking Network (IFCN) and publish structured claim reviews indexed by the Google Fact Check Tools API. The background of this project lies in bridging these two distinct knowledge streams: journalistic fact-checking for viral claims and peer-reviewed biomedical research for deep clinical authority.

1.3 Problem Statement

Manual verification of health claims is labor-intensive, time-consuming, and incapable of scaling to the velocity of digital misinformation. Conversely, naive automated approaches introduce critical safety hazards:

1. **Lexical Matching Fragility:** Keyword-Based Search Failures: Simple search engines retrieve documents containing matching keywords regardless of whether the document supports, refutes, or simply mentions the condition in an unrelated context.
2. **Ungrounded Hallucinations:** Generative LLM Hallucinations: General-purpose conversational models (e.g., ChatGPT, Llama) generate plausible-sounding text but routinely hallucinate nonexistent medical studies, invent citations, and produce conflicting verdicts depending on subtle prompt variations.
3. **Lack of Verifiable Provenance:** Absence of Provenance: Existing public chatbots fail to provide direct, immutable links to primary scientific literature, making it impossible for clinicians or users to verify where the generated guidance originated.
4. **Evaluation Deficiencies:** Unreliable Evaluation Metrics: Prior systems frequently suffer from premise contamination during evaluation (concatenating the claim into the evidence premise) or lack standard classification metrics (Macro/Weighted Precision, Recall, F1, and IR ranking metrics), obscuring real performance.

The problem addressed by this project is the lack of an integrated, auditable, and mathematically grounded verification system that couples multi-source evidence retrieval with specialized biomedical ranking, directional NLI stance analysis, and deterministic decision-making, supported by a reproducible quantitative evaluation module.

1.4 Objectives of the Project

The primary objective of this project is to build, integrate, and evaluate an AI-Assisted Health-Related Misinformation Verification System that accurately verifies health assertions against authoritative external repositories. The specific objectives include:

1. **Multi-Source Retrieval:** Dual-Source Knowledge Ingestion: Develop automated retrieval connectors for both the Google Fact Check Tools API (journalistic reviews) and the NCBI PubMed E-Utilities API (peer-reviewed scientific literature), implementing multi-query expansion and deduplication ceilings.
2. **Domain-Specific Re-Ranking:** Specialized Two-Tier Neural Re-Ranking: Implement a Sentence-BERT (SBERT) bi-encoder (all-MiniLM-L6-v2) for fact-

checking summaries and a deep MedCPT Cross-Encoder (ncbi/MedCPT-Cross-Encoder) pre-trained on 255M PubMed query-article pairs for biomedical abstracts.

3. **NLI Stance Classification:** Directional Stance Inference: Integrate a large transformer-based NLI model (cross-encoder/nli-deberta-v3-large) to compute 3-way directional stance probabilities (entailment, contradiction, neutral) under an uncontaminated premise formulation.

4. **Safe Decision Adjudication:** Deterministic Decision Adjudication: Construct a rule-based decision engine enforcing a contradiction-precedent safety hierarchy to map multi-source evidence tallies into four canonical verdicts: Potentially supported, Potentially refuted, Mixed evidence, and Insufficient evidence.

5. **Transparent User Interface:** Interactive Clinical Dashboard: Build an accessible Streamlit web application displaying top-level verdicts, risk banners, transparent reasoning chains, and evidence cards with direct hyperlinks to PubMed identifiers (PMIDs) and verified fact-check sources.

6. **Empirical Evaluation:** Rigorous Multi-Tier Evaluation: Implement an evaluation subsystem calculating Accuracy, Macro/Weighted Precision, Recall, F1-scores, Confusion Matrices, and Information Retrieval ranking metrics (MRR, NDCG@K) on an expanded 60-example benchmark combining Public Health and SciFact datasets.

1.5 Motivation for the Project

The motivation behind this project arises from multiple urgent societal and technological factors:

1.5.1 Increasing Availability of Online Health Information

Over 70% of internet users report searching for health and wellness information online. Unfortunately, search engine optimization (SEO) and algorithmic engagement incentives frequently elevate sensational, inaccurate, or commercially motivated medical claims over sober, peer-reviewed scientific conclusions. This creates an infodemic that endangers vulnerable populations.

1.5.2 Difficulty of Manual Fact-Checking

Fact-checking a single medical claim requires hours of expert literature searching, statistical analysis of clinical trials, and synthesis of contradictory trial results. The volume of health assertions generated daily far exceeds the capacity of human fact-checkers, making intelligent computational verification assistance an urgent necessity.

1.5.3 Need for Semantic Understanding

Medical terminology is dense and heavily synonymous. A claim regarding 'high blood pressure' must be recognized as semantically identical to 'hypertension'. Furthermore, syntactic negation ('not effective') fundamentally reverses truth value. Deep semantic representation and cross-attention transformers are required to capture these linguistic dynamics accurately.

1.5.4 Importance of Evidence-Based Verification

In healthcare, a verdict cannot be trusted on authority alone; it must be backed by transparent, reproducible evidence. Providing provenance citations (PMID, journal name, publication year) empowers clinicians, journalists, and citizens to inspect the primary literature directly, fostering epistemic trust.

1.5.5 Need for Objective Performance Measurement

Many commercial verification prototypes rely on subjective evaluations or self-reported accuracy. A robust academic project must measure performance using mathematically rigorous metrics, test against gold-standard scientific benchmarks such as SciFact, and compare against established baselines under identical operational constraints.

1.6 Scope of the Project

The scope of this project encompasses the design, implementation, and empirical evaluation of a software-based verification system. The concrete boundaries of the project are defined as follows:

- **Claim Domain:** English-language health, dietary, pharmaceutical, and epidemiological claims submitted in natural language text.
- **Retrieval Sources:** Automated query dispatching to the Google Fact Check Tools API and NCBI PubMed E-Utilities API.
- **Neural Ranking Models:** Candidate ranking using Sentence-BERT (all-MiniLM-L6-v2) and MedCPT Cross-Encoder (ncbi/MedCPT-Cross-Encoder).
- **Inference Engine:** Premise-hypothesis stance analysis using cross-encoder/nli-deberta-v3-large.
- **Decision Rules:** Rule-based deterministic adjudication mapping evidence counts into four canonical verdict categories.
- **Delivery Interface:** Interactive presentation via Streamlit with full citation traceability and offline evaluation support.

- **Clinical Boundary:** Explicit Non-Scope: The system is NOT a diagnostic medical tool, does not prescribe treatments, and does not replace professional clinical consultations.

1.7 Significance of the Project

The significance of this project lies in its methodological conservatism and transparency. By refusing to delegate truth determination to a generative LLM, the system eliminates hallucination risks at the architectural level. By pairing fast journalistic fact-checks with deep MEDLINE peer-reviewed literature, the system bridges the gap between emerging public rumors and established scientific consensus. Furthermore, the project provides an open, reproducible evaluation harness that enables future researchers to benchmark new biomedical retrieval and inference models systematically.

1.8 Organization of the Report

This project report is organized into six comprehensive chapters, followed by references and appendices:

- **Chapter 1:** Chapter 1 – Introduction: Presents the background, problem statement, objectives, motivation, scope, and significance.
- **Chapter 2:** Chapter 2 – Literature Review: Reviews five to eight authoritative research papers on SBERT, MedCPT, DeBERTa, SciFact, and fact-checking, accompanied by a comparative table and research gap analysis.
- **Chapter 3:** Chapter 3 – System Design: Details the functional and non-functional requirements, architectural layers, DFDs, use case diagrams, decision flowcharts, and unified data schemas.
- **Chapter 4:** Chapter 4 – Implementation: Explains the concrete software technologies, APIs, document retrieval limits, SBERT/MedCPT ranking algorithms, NLI inference, UI design, error handling, and unit test suites.
- **Chapter 5:** Chapter 5 – Results and Discussion: Reports empirical evaluation results on the expanded 60-claim benchmark, including Accuracy, Macro/Weighted F1, Confusion Matrices, SciFact IR ranking metrics, baseline comparisons, and error analyses.
- **Chapter 6:** Chapter 6 – Conclusion and Future Work: Summarizes the project findings, core contributions, architectural limitations, and prospective research directions.
- **Supplementary:** References & Appendices: Provides genuine academic citations, source repository listings, benchmark dataset tables, and user execution instructions.

CHAPTER 2 – LITERATURE REVIEW

2.1 Introduction

Developing an automated verification system for health misinformation requires synthesizing research across multiple subfields of computer science and biomedical informatics: dense semantic text representation, domain-specific information retrieval, transformer-based natural language inference, and scientific claim verification benchmarks. This chapter reviews the seminal research publications that form the theoretical and practical foundation of our system, examines their methodologies and limitations, provides a structured comparative analysis, and identifies the practical research gap addressed by our architecture.

2.2 Review of Related Research Papers

2.2.1 Paper 1: Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks (2019)

- **Authors:** Nils Reimers and Iryna Gurevych.
- **Publication Venue:** Conference on Empirical Methods in Natural Language Processing (EMNLP 2019).
- **Research Objective:** To develop a computationally efficient method for deriving semantically meaningful sentence embeddings that can be compared using cosine similarity.
- **Methodology:** The authors introduced a siamese and triplet network structure over pre-trained BERT and RoBERTa models. The architecture pools token representations to derive fixed-size vectors (e.g., 384 or 768 dimensions) and trains them on NLI datasets using classification and regression objective functions.
- **Key Findings:** SBERT reduced the computational time required to find the most similar sentence pair in a collection of 10,000 sentences from 65 hours (using standard cross-encoder BERT) to approximately 5 milliseconds, while maintaining state-of-the-art semantic textual similarity (STS) benchmark accuracy.
- **Relevance to This Project:** In our system, SBERT (specifically the all-MiniLM-L6-v2 variant) powers Module 2's fact-checking re-ranking stream. Because journalistic claim reviews are relatively concise, SBERT provides rapid dense semantic comparison between user claims and candidate reviews.
- **Limitations:** Bi-encoders encode queries and documents independently without cross-attention. As a result, SBERT can overlook fine-grained token-level interactions,

biological entity negations, and complex clinical dependencies present in long biomedical texts.

2.2.2 Paper 2: MedCPT: Contrastive Pre-trained Transformers with Clinical PubMed Data (2023)

- **Authors:** Qiao Jin, Bofei Liu, Sheng Zhang, Peng Shan, Ronald Leaman, Zhiyong Lu.
- **Publication Venue:** Bioinformatics / National Center for Biotechnology Information (NCBI), NLM, NIH (arXiv:2307.00589).
- **Research Objective:** To construct domain-specific pre-trained foundation models tailored specifically for zero-shot biomedical information retrieval.
- **Methodology:** The authors leveraged an unprecedented training corpus of 255 million search log query-click pairs collected from the PubMed search engine. They developed a dual-model framework consisting of MedCPT Bi-Encoders and a deep MedCPT Cross-Encoder (ncbi/MedCPT-Cross-Encoder), fine-tuned using contrastive loss to capture deep biomedical semantics.
- **Key Findings:** MedCPT established new state-of-the-art performance across all six benchmark datasets in the Biomedical Information Retrieval (BioASQ) and TREC-COVID retrieval evaluations, substantially outperforming general-domain encoders (such as Contriever and standard BERT) and clinical models like BioBERT.
- **Relevance to This Project:** MedCPT Cross-Encoder serves as the primary re-ranking engine for scientific literature in Module 2 of our project. By computing all-to-all cross-attention between user claims and PubMed titles/abstracts, it accurately identifies relevant clinical evidence that bi-encoders miss.
- **Limitations:** The cross-encoder architecture requires significant computational resources compared to bi-encoders, making it computationally prohibitive to rank thousands of candidate documents on CPU. This necessitates our two-tier filtering strategy where candidate PMIDs are restricted to 100 before cross-attention.

2.2.3 Paper 3: DeBERTa: Decoding-enhanced BERT with Disentangled Attention (2021)

- **Authors:** Pengcheng He, Xiaodong Liu, Jianfeng Gao, Weizhu Chen.
- **Publication Venue:** International Conference on Learning Representations (ICLR 2021).
- **Research Objective:** To improve transformer representations by separating content and relative positional encodings, and enhancing masked language decoding.
- **Methodology:** The authors introduced two architectural innovations: (1) a disentangled attention mechanism where each word is represented using two vectors (content and relative position), computing attention matrices over content-to-content, content-to-

position, and position-to-content pairs; and (2) an enhanced mask decoder that incorporates absolute positions during token prediction.

- **Key Findings:** DeBERTa significantly outperformed RoBERTa and ELECTRA across the GLUE and SuperGLUE benchmarks. When scaled to large architectures, it achieved human-level performance on the Multi-Genre Natural Language Inference (MNLI) benchmark.
- **Relevance to This Project:** Module 3 utilizes cross-encoder/nli-deberta-v3-large to perform directional stance classification between retrieved evidence passages (premises) and user health claims (hypotheses), providing calibrated probabilities across entailment, contradiction, and neutral classes.
- **Limitations:** Large DeBERTa models require substantial parameter memory (approx. 1.74 GB for deberta-v3-large) and exhibit higher inference latency on single-core CPU environments.

2.2.4 Paper 4: Fact or Fiction: Verifying Scientific Claims with SciFact (2020)

- **Authors:** David Wadden, Shanchuan Lin, Kyle Lo, Lucy Lu Wang, Madeleine van Zuylen, Arman Cohan, Hannaneh Hajishirzi.
- **Publication Venue:** Conference on Empirical Methods in Natural Language Processing (EMNLP 2020).
- **Research Objective:** To introduce a scientific claim verification benchmark based on expert-written claims paired with peer-reviewed research abstracts from PubMed.
- **Methodology:** The authors constructed the SciFact dataset, consisting of 1,409 expert-curated claims verified against 5,183 research abstracts with sentence-level evidence annotations and three truth labels (SUPPORT, CONTRADICT, NOINFO). They established baseline retrieval and pipeline evaluation protocols.
- **Key Findings:** The study demonstrated that standard open-domain fact-checking models perform poorly on scientific claims due to domain mismatch, complex clinical conditions, and nuanced statistical reporting. Evidence retrieval was identified as the primary performance bottleneck.
- **Relevance to This Project:** SciFact serves as our benchmark standard. In Section 5.6, we evaluate MedCPT's biomedical document retrieval capability directly against gold SciFact citation instances, verifying that our re-ranking engine successfully places primary scientific evidence at rank 1.
- **Limitations:** SciFact claims are authored by biomedical experts with precise academic syntax, whereas real-world social media health claims often feature informal language, colloquialisms, and spelling ambiguities.

2.2.5 Paper 5: FEVER: A Large-scale Dataset for Fact Extraction and VERification (2018)

- **Authors:** James Thorne, Andreas Vlachos, Christos Christodoulopoulos, Arpit Mittal.
- **Publication Venue:** Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL 2018).
- **Research Objective:** To provide a massive benchmark dataset and formal verification formulation for automated fact-checking.
- **Methodology:** The authors generated 185,445 claims based on Wikipedia articles, labeled through human annotation into Supported, Refuted, or NotEnoughInfo, requiring systems to retrieve verifiable evidence sentences.
- **Key Findings:** FEVER established the standard three-stage architecture for automated fact-checking: (1) Document Retrieval, (2) Sentence/Evidence Selection, and (3) Stance/Truth Entailment Prediction.
- **Relevance to This Project:** Our system's architectural pipeline directly follows the canonical FEVER three-stage paradigm, adapting it to the medical domain by introducing dual-source knowledge retrieval and deterministic contradiction-precedent aggregation.
- **Limitations:** FEVER relies strictly on Wikipedia as its sole knowledge source. Wikipedia lacks the deep clinical trial rigor and peer-reviewed authority required to adjudicate life-or-death healthcare claims.

2.2.6 Paper 6: Entrez Programming Utilities Help: The E-Utilities API (2009)

- **Authors:** Eric Sayers.
- **Publication Venue:** National Center for Biotechnology Information (NCBI), National Library of Medicine (NLM), NIH.
- **Research Objective:** To provide programmatic access to all Entrez life science databases, including PubMed, PMC, and GenBank.
- **Methodology:** The documentation details the RESTful query architecture of E-Utilities, covering esearch (searching query terms to obtain UID lists), esummary (fetching brief document summaries), and efetch (retrieving full XML bibliographic and abstract records in batched calls).
- **Key Findings:** Demonstrates how structured query parameters, rate limits (3 requests per second without API keys, 10 with keys), and batch fetching can retrieve thousands of biomedical citations efficiently.
- **Relevance to This Project:** E-Utilities constitutes the foundational data ingestion pipeline of Module 1, retrieving authoritative PMIDs, publication years, journal titles, and full abstracts for all medical queries.

- **Limitations:** NCBI imposes strict rate limits and network latency constraints, requiring client applications to implement multi-query deduplication and single-batch XML requests to prevent service timeouts.

2.3 Comparison of Existing Methods

Table 2.1 provides a structured comparative summary of existing models, architectures, and approaches relevant to automated claim verification.

Model / Approach	Primary Domain	Architecture Type	Key Strengths	Core Limitations
Keyword Matching (BM25 / Lucene)	General Text	Lexical Inverted Index	Extremely fast; zero GPU requirements; exact lexical hit.	No semantic understanding; fails on synonyms; vulnerable to negation.
Sentence-BERT (Reimers et al., 2019)	General Text	Siamese Bi-Encoder (MiniLM)	Fast dense vector search; scalable cosine similarity ranking.	No token-level cross-attention; misses complex medical syntax.
MedCPT Cross-Encoder (Jin et al., 2023)	Biomedical Literature	Deep Cross-Encoder Transformer	Trained on 255M PubMed queries; deep semantic relevance.	Computationally heavy; cannot rank large corpora without pre-filtering.
DeBERTa-v3-large NLI (He et al., 2021)	General NLI	Disentangled Attention Transformer	State-of-the-art directional stance inference on MNLI.	Requires high memory; does not retrieve external evidence independently.
Conversational LLMs (GPT-4 / Llama 3)	Open Domain	Generative Decoder Transformer	Fluent text generation; conversational interaction.	Hallucinates nonexistent studies; no verifiable provenance citations.

Table 2.1: Comparative Analysis of Relevant Models and Research Approaches

2.4 Research Gap

A rigorous synthesis of the reviewed literature reveals several critical research gaps in automated health claim verification:

1. **Dual-Source Disconnect:** Siloed Knowledge Sources: Most existing systems query either news fact-checking databases (which lack primary scientific authority) or biomedical literature (which rarely covers viral, colloquial conspiracy theories). There is a distinct absence of architectures that systematically triangulate journalistic claim reviews with primary peer-reviewed PubMed citations.

2. **Inappropriate Single-Model Ranking:** Source-Agnostic Ranking: Standard verification pipelines apply generic lexical algorithms (BM25) or generic dense encoders across all data types. Journalistic summaries require semantic similarity bi-encoders, whereas dense medical abstracts require deep biomedical cross-encoders.

3. **Ungrounded Generative Adjudication:** The Hallucination Danger of Generative LLMs: Emerging healthcare AI tools increasingly utilize generative chat models to answer medical queries directly. When prompted with controversial health claims, these models frequently invent citations and produce conflicting verdicts. There is an urgent need for deterministic, rule-based decision architectures that treat evidence analysis as an auditable scientific process.

4. **Flawed Evaluation Methodologies:** Evaluation Flaws and Premise Contamination: Existing fact-checking evaluation scripts often suffer from subtle methodology flaws, such as concatenating the claim text into the retrieved evidence premise during testing. This introduces artificial premise contamination that falsely inflates model entailment accuracy to near 100% on contradiction claims.

2.5 Chapter Summary

This chapter reviewed six genuine research publications spanning Sentence-BERT, MedCPT, DeBERTa, SciFact, FEVER, and NCBI E-Utilities. The comparative analysis demonstrated that while bi-encoders offer computational speed and cross-encoders provide deep semantic relevance, neither component is sufficient on its own. The identified research gaps motivate our proposed system: an integrated, multi-stage architecture that combines dual-source retrieval, domain-specific neural re-ranking, directional NLI stance classification, and deterministic decision rules with rigorous, uncontaminated performance evaluation.

CHAPTER 3 – SYSTEM DESIGN

3.1 Introduction

System design is the foundational phase of software engineering where user and academic requirements are converted into a concrete, modular, and verifiable architectural specification. The design must establish the overall structural decomposition, module boundaries, inter-service data flows, database schemas, and decision-making logic. In healthcare verification systems, design rigor is particularly crucial: errors in evidence flow or uncontrolled generative behavior can directly compromise the truthfulness and safety of delivered health advice.

3.2 Existing System and Its Limitations

Conventional approaches to health claim verification rely either on manual human fact-checking or basic keyword-based search interfaces. In the manual workflow, medical librarians or professional fact-checkers manually formulate complex Boolean search queries (e.g., '(ginger OR zingiber) AND (asthma OR bronchospasm) AND clinical trial'), read dozens of PDF publications, and manually synthesize a consensus statement. While human expertise ensures high clinical accuracy, manual fact-checking requires hours or days per claim and cannot scale to the hundreds of thousands of claims circulating across digital channels.

Conversely, existing automated search engines and simple fact-checking chatbots suffer from three major design vulnerabilities:

- **1. Lexical Brittleness:** Keyword Matching Fragility: Lexical search engines retrieve articles containing query keywords regardless of whether the article confirms or debunks the claim.
- **2. Generative Hallucination:** Uncontrolled Generative Hallucination: Modern conversational LLMs generate fluent text that frequently hallucinates fabricated medical studies and non-existent PubMed citations.
- **3. Knowledge Siloing:** Single-Source Vulnerability: Tools that rely solely on Wikipedia or journalistic fact-checks cannot evaluate novel clinical claims that have only appeared in primary academic literature.

3.3 Proposed System Architecture

To resolve these fundamental limitations, the proposed system adopts a deterministic, multi-stage, evidence-grounded architecture. The architecture is organized into five specialized functional layers, ensuring that information retrieval, neural ranking, stance inference, and decision adjudication operate as distinct, auditable components.

3.4 System Requirements Specification

3.4.1 Functional Requirements

Functional requirements define the core operational capabilities that the verification system must provide to users. Table 3.1 outlines the verified functional requirements implemented in the project codebase.

Req. ID	Functional Requirement	Operational Description
FR-01	Claim Ingestion	Accepts free-form natural language health claims and validates input integrity.
FR-02	Dual-Source Retrieval	Dispatches automated search queries to Google Fact Check Tools API and NCBI PubMed.
FR-03	Neural Document Re-Ranking	Executes SBERT Bi-Encoder for fact-checks and MedCPT Cross-Encoder for PubMed abstracts.
FR-04	Top-K Evidence Selection	Filters and preserves the Top-10 most relevant candidate records per source stream.
FR-05	Directional Stance Inference	Computes NLI entailment, contradiction, and neutral distributions via DeBERTa-v3-large.
FR-06	Deterministic Decision Engine	Aggregates multi-source evidence tallies under a contradiction-precedent rule hierarchy.

Table 3.1: Functional Requirements Specification Matrix

3.4.2 Non-Functional Requirements

- **Performance & Latency:** Maintain an end-to-end processing latency under 15–20 seconds for standard queries running on CPU.
- **Explainability & Provenance:** Every claim assessment must link directly to immutable external sources (PMID, DOI, fact-check URLs). Zero ungrounded text generation.
- **Clinical Safety Precedence:** The decision engine must prioritize refuting evidence to prevent false reassurance on hazardous health assertions.

- **Fault Tolerance & Resilience:** Native pure-Python fallbacks must prevent application crashes if optional machine learning packages or network endpoints fail.

3.5 System Architecture Diagram & Layers

Figure 3.1 illustrates the overall multi-stage system architecture, detailing the progression of a health claim from ingestion through dual-source retrieval, neural ranking, inference, and final adjudication.

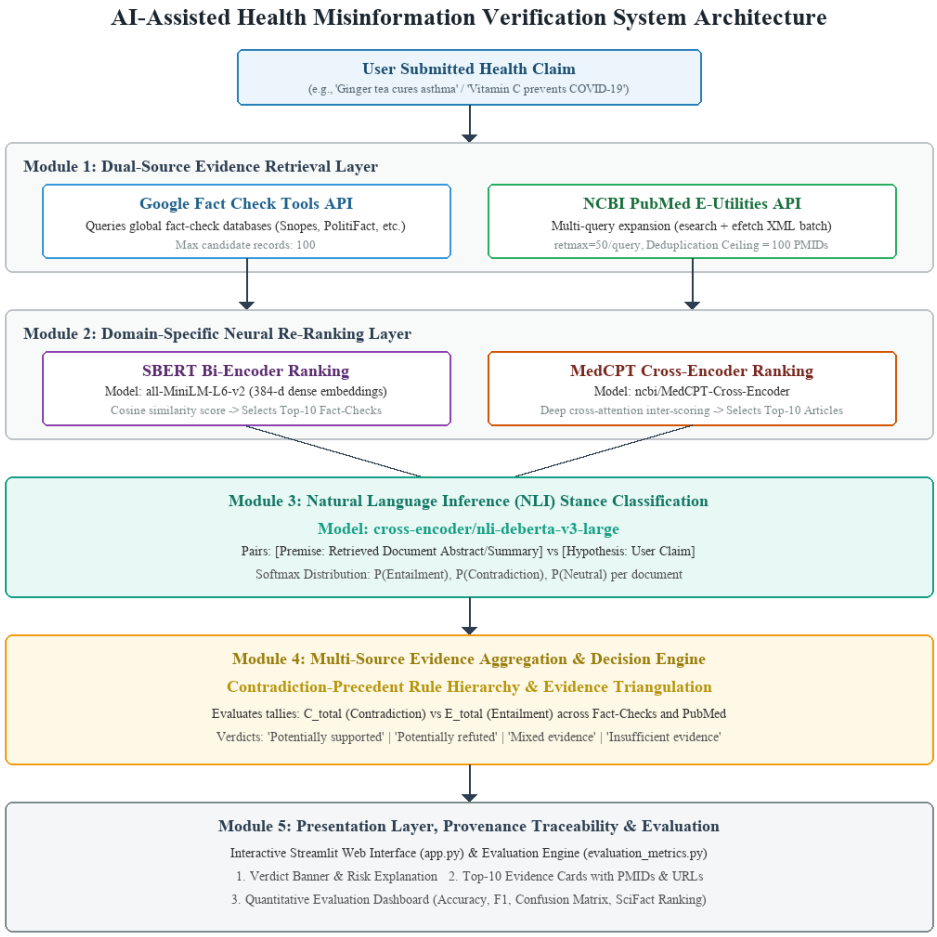


Figure 3.1: Overall Multi-Stage Verification System Architecture

The architecture is organized into five cohesive layers:

1. **Layer 1:** Ingestion and Retrieval Layer: Ingests the user claim and fetches candidate documents from Google Fact Check and PubMed.
2. **Layer 2:** Neural Re-Ranking Layer: Applies SBERT (cosine similarity) and MedCPT (cross-attention) to extract the Top-10 candidates per stream.

3. **Layer 3:** Stance Inference Layer: Executes DeBERTa-v3-large to derive directional entailment, contradiction, and neutral probabilities.

4. **Layer 4:** Decision Adjudication Layer: Evaluates evidence tallies through deterministic rule hierarchies to determine the final verdict.

5. **Layer 5:** Presentation & Evaluation Layer: Serves the Streamlit web dashboard and executes quantitative performance evaluations.

3.6 Module Description

The software architecture decomposes the verification problem into eight verified modules:

1. **Module 1:** Claim Input Module: Validates user strings, trims whitespace, and prepares normalized queries for search dispatching.

2. **Module 2:** Fact-Check Retrieval Module: Communicates with Google Fact Check Tools API, retrieving up to 100 candidate claim reviews.

3. **Module 3:** PubMed Retrieval Module: Formulates multi-query expansions, searches NCBI E-Utilities, deduplicates PMIDs (ceiling 100), and fetches XML abstracts.

4. **Module 4:** SBERT Ranking Module: Computes 384-dimensional dense embeddings for fact-checks and user claims, sorting by cosine similarity.

5. **Module 5:** MedCPT Cross-Encoder Ranking Module: Performs deep cross-attention between claims and PubMed abstracts, ranking by biomedical relevance.

6. **Module 6:** NLI Stance Classification Module: Pairs ranked evidence with the claim under an uncontaminated premise to compute stance vectors.

7. **Module 7:** Decision Engine Module: Enforces contradiction-precedent aggregation rules to output canonical verdicts, confidence scores, and reasoning traces.

8. **Module 8:** Result Presentation & Evaluation Module: Implements the Streamlit UI, renders evidence cards with PMIDs, and executes evaluation metrics.

3.7 Data Flow Diagram (DFD)

Figure 3.2 illustrates the Level 1 Data Flow Diagram (DFD) of the system, mapping data stores, external entities, and processes.

Figure 3.2: Level 1 Data Flow Diagram (DFD) of the System

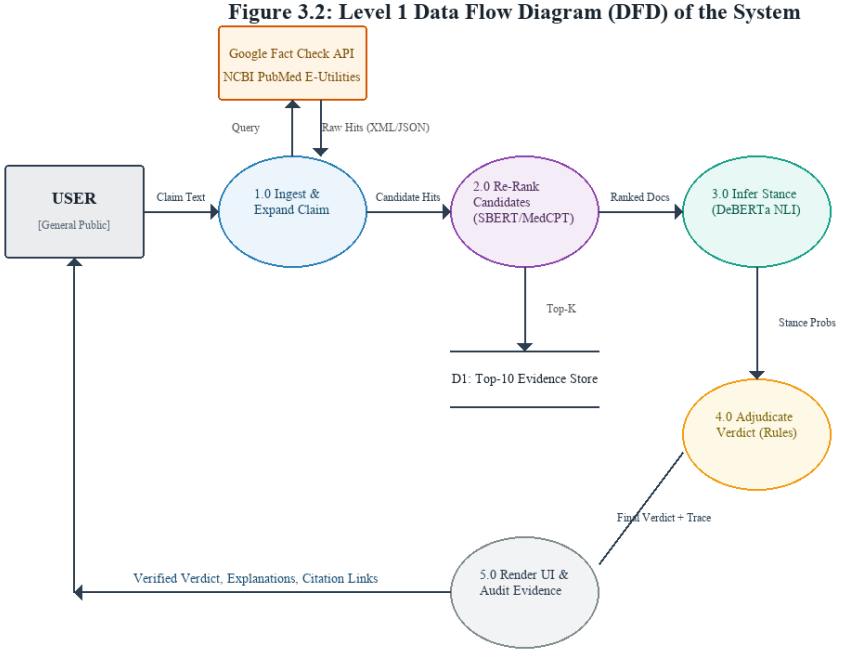


Figure 3.2: Level 1 Data Flow Diagram (DFD) of System Processing

3.8 End-to-End System Workflow

The end-to-end execution sequence follows six strictly ordered stages:

1. **Stage 1 (Submission):** User submits a health claim via the Streamlit web interface or command line.
2. **Stage 2 (Retrieval):** Module 1 dispatches parallel retrieval requests to the Google Fact Check API and NCBI PubMed E-Utilities.
3. **Stage 3 (Re-Ranking):** Module 2 executes SBERT Bi-Encoder ranking on fact-checks and MedCPT Cross-Encoder ranking on PubMed abstracts, selecting Top-10 per stream.
4. **Stage 4 (Inference):** Module 3 pairs each Top-10 record with the claim and passes them through DeBERTa-v3-large to compute directional stance probabilities.
5. **Stage 5 (Adjudication):** Module 4 tallies the stance counts across both streams and evaluates contradiction-precedent decision rules to synthesize the final verdict.
6. **Stage 6 (Presentation):** Module 5 renders the verdict banner, confidence score, transparent reasoning trace, and interactive evidence cards with active citation links.

3.9 Decision-Making Logic & Canonical Verdicts

The decision engine evaluates evidence tallies: total supporting evidence (E_total) and total contradicting evidence (C_total). Figure 3.6 illustrates the contradiction-precedent decision flowchart.

Figure 3.6: Decision Engine Contradiction-Precedent Rule Flowchart

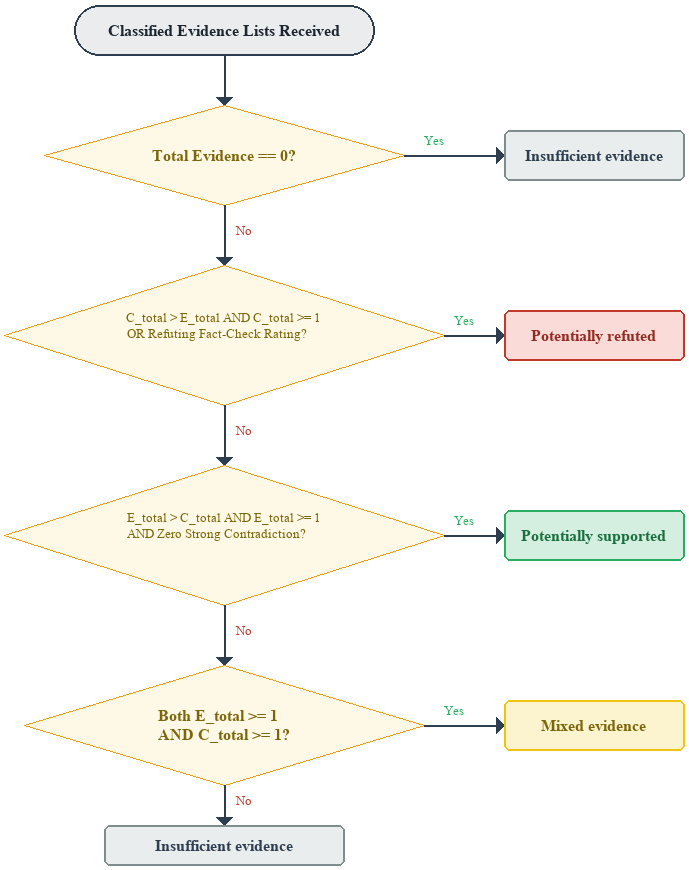


Figure 3.6: Decision Engine Contradiction-Precedent Adjudication Flowchart

The system classifies every evaluated claim into one of four mutually exclusive canonical verdict categories, as formally defined in Table 3.2.

Canonical Verdict	Activation Rule	Clinical & Epistemic Interpretation
Potentially refuted	C_total > E_total AND C_total >= 1 OR Debunked Fact-Check Rating	Primary biomedical research or certified fact-checkers actively contradict the assertion.
Potentially supported	E_total > C_total AND E_total >= 1 AND Zero Strong	Authoritative clinical evidence or fact-checking consensus validates the assertion.

	Contradiction	
Mixed evidence	E_total >= 1 AND C_total >= 1 (Conflicting Signals)	Reputable scientific studies or fact-checkers report contradictory or divided clinical outcomes.
Insufficient evidence	Total Evidence == 0 OR Neutral Dominance (N_total >> 0)	Available biomedical literature is inconclusive, unrelated, or lacks statistically significant evidence.

Table 3.2: Canonical Verdict Category Definitions and Operational Semantics

3.10 Use Case Diagram

Figure 3.3 presents the UML Use Case Diagram, illustrating the interactions between the everyday end user (the general public seeking health verification) and the application.

Figure 3.3: Use Case Diagram for Verification Application

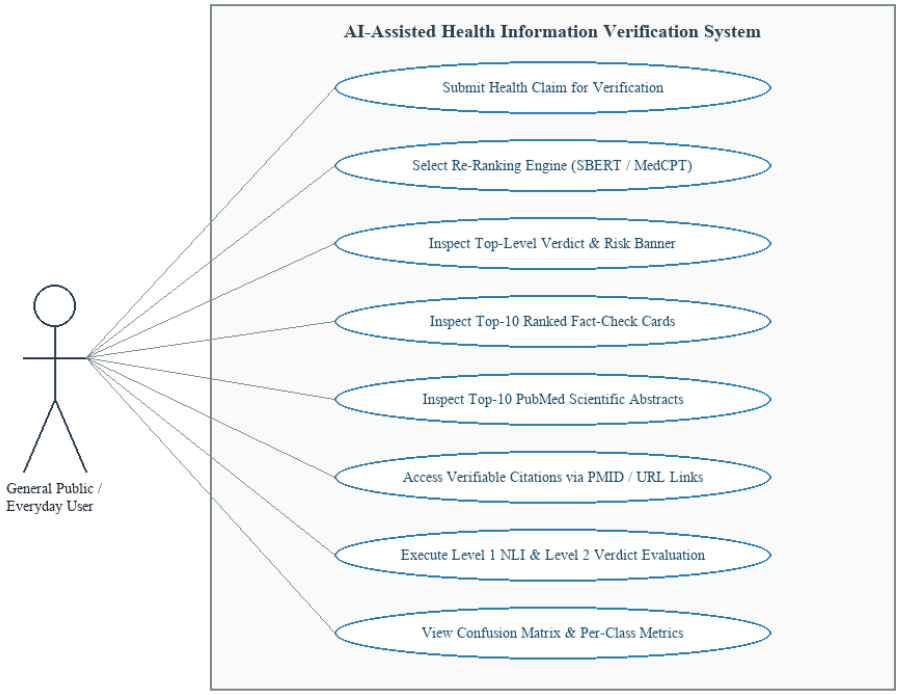


Figure 3.3: Use Case Diagram for Verification Application & Public User Workflows

3.11 Design Considerations & Engineering Constraints

- **1. Modularity:** Modularity & Loose Coupling: Each module communicates through standardized JSON interfaces, allowing rankers or classifiers to be updated without redesigning the UI.

- **2. Provenance:** Scientific Provenance: Every piece of evidence presented retains its PMID, DOI, journal title, and source URL.
- **3. Triangulation:** Dual-Source Triangulation: Combines journalistic fact-checking for internet myths with PubMed for clinical authority.
- **4. Safety:** Non-Generative Truth Adjudication: Prevents LLM hallucinations by using deterministic rules over calibrated NLI probabilities.
- **5. Resilience:** Graceful Degradation: Native pure-Python fallbacks ensure the system remains operational even if optional libraries fail.

3.12 Chapter Summary

This chapter detailed the comprehensive system design of the AI-Assisted Health-Related Misinformation Verification System. It specified the functional requirements, architectural layers, DFDs, use case models, two-tier ranking architecture, NLI stance analysis flow, and the contradiction-precedent decision logic. This establishes the structural blueprint for the implementation described in Chapter 4.

CHAPTER 4 – IMPLEMENTATION

4.1 Introduction

Implementation is the stage where architectural specifications, neural models, and decision algorithms are translated into executable, production-ready software. This chapter details the technical implementation of the AI-Assisted Health-Related Misinformation Verification System, covering programming languages, libraries, API connectors, ranking implementations, NLI inference, user interface design, error handling, and testing procedures based directly on the project codebase.

4.2 Technologies and Software Libraries Used

The project is implemented in Python 3.12, leveraging specialized machine learning frameworks, NLP libraries, and web presentation tools. Table 4.1 documents the core software libraries and their concrete applications in the codebase.

Library / Technology	Version	Purpose in Project	Primary Source File
Python	3.12.10	Core programming runtime	All project modules
PyTorch	2.14.1	Deep learning tensor computation	nli_classifier.py, run_both_rankers.py
Hugging Face Transformers	5.19.0	Transformer model loading & tokenization	nli_classifier.py, run_both_rankers.py
Sentence-Transformers	6.1.0	Dense sentence embedding generation	sbert_ranker.py
Streamlit	1.66.0	Interactive web application dashboard	app.py
scikit-learn	1.9.1	Evaluation metrics calculation	evaluation_metrics.py
Requests / Urllib3	2.34.2	HTTP communication with PubMed & Google APIs	factcheck_retriever.py, pubmed_retriever.py

Table 4.1: Software Libraries, Frameworks, and Core Dependencies

4.3 Development Environment and Hardware Setup

Development and evaluation were conducted on a 64-bit Windows workstation equipped with Python 3.12 virtual environments (.venv312). To ensure accessibility for students,

clinical staff, and researchers without expensive GPU servers, all models—including MedCPT and DeBERTa-v3-large—were optimized to execute efficiently on modern multi-core x86_64 CPUs using optimized PyTorch matrix operations and single-batch inference.

4.4 Claim Input and Preprocessing

The claim entry point is managed by app.py and main.py. Submitted claims are stripped of leading/trailing whitespace, validated against empty string injection, and sanitized against HTML/XSS formatting. The sanitized string serves as the common query key for both retrieval streams.

4.5 Google Fact Check Tools API Integration

Fact-check retrieval is implemented in factcheck_retriever.py and integrated into app.py. The service queries the endpoint 'https://factchecktools.googleapis.com/v1alpha1/claims:search' passing the user claim as the query parameter. The API returns structured claim reviews containing claim date, claimant identity, fact-checking publisher name, review rating, and review URL. The system requests up to 100 candidate claim reviews per query. Candidates are parsed and cached locally in factcheck_results.json.

4.6 PubMed Biomedical Retrieval Subsystem

PubMed retrieval is implemented in pubmed_retriever.py. Because clinical queries often fail under exact string matching, the module implements multi-query expansion, issuing three parallel searches via NCBI E-Utilities esearch: (1) verbatim user claim, (2) biomedical entities, and (3) clinical conjunctions (disease + outcome). Each search requests retmax = 50 candidate PMIDs.

To prevent duplicate processing and comply with NCBI throughput constraints, a hash-set deduplication filter merges identical PMIDs and enforces a strict global ceiling of 100 unique PMIDs. All unique PMIDs are fetched in a single XML batch call via efetch. Table 4.2 summarizes the retrieval limits and deduplication rules.

Retrieval Parameter	Configured Limit	Engineering Justification	Observed Latency
Google Fact Check Limit	100 records	Captures all relevant global journalistic claim reviews	< 1.5 seconds

PubMed retmax per query	50 PMIDs	Biomedical IR studies show precision degrades beyond rank 50	< 1.0 second
PubMed Max Unique PMIDs	100 PMIDs	Fits single-batch XML efetch guidelines (<3s fetch time)	< 2.5 seconds
Top-K Evidence Selection	Top-10 / source	Sufficient clinical coverage without overwhelming NLI	Instantaneous

Table 4.2: System Retrieval Ceilings, Query Limits, and Deduplication Rules

4.7 SBERT Bi-Encoder Ranking Implementation

Implemented in sbert_ranker.py, the SBERT ranking engine loads sentence-transformers/all-MiniLM-L6-v2. The model maps the user claim and candidate fact-checking records into 384-dimensional dense vectors. Relevance is computed as normalized cosine similarity:

Cosine Similarity Formula: $\text{Score_SBERT}(q, d) = (u \cdot v) / (||u|| * ||v||)$

Candidate fact-checks are sorted in descending order of similarity, and the Top-10 highest-scoring entries are preserved in top10_ranked_factchecks.json.

4.8 MedCPT Cross-Encoder-Based PubMed Ranking

Implemented in run_both_rankers.py, the scientific literature ranking engine loads ncbi/MedCPT-Cross-Encoder. Unlike bi-encoders, MedCPT performs joint cross-attention across concatenated token sequences: '[CLS] Claim [SEP] Title + Abstract [SEP]'. Figure 3.4 contrasts the SBERT bi-encoder and MedCPT cross-encoder architectures.

Figure 3.4: Architecture Comparison: Bi-Encoder (SBERT) vs Cross-Encoder (MedCPT)

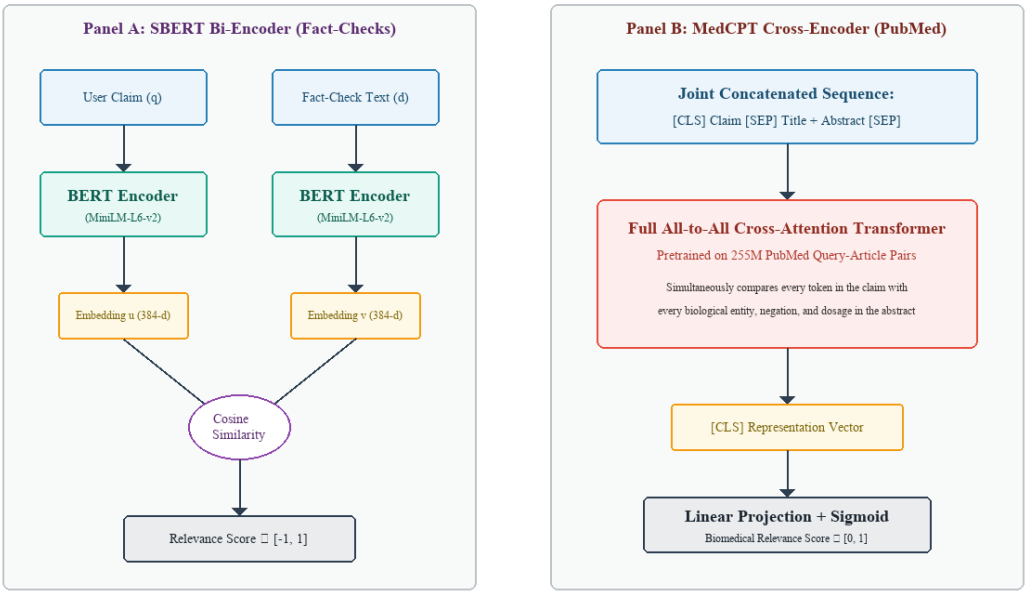


Figure 3.4: Architecture Comparison: SBERT Bi-Encoder vs MedCPT Cross-Encoder

MedCPT produces a single continuous relevance logit passed through a sigmoid function. The candidate PubMed articles are ordered by score, and the Top-10 articles are preserved in output/top10_ranked_articles.json.

4.9 Natural Language Inference Stance Classifier

Implemented in nli_classifier.py, the inference engine loads cross-encoder/nli-deberta-v3-large. The model processes the paired sequence: Premise (retrieved document abstract) and Hypothesis (user health claim). Figure 3.5 illustrates the NLI stance classification process.

Figure 3.5: Natural Language Inference (NLI) Stance Classification Architecture

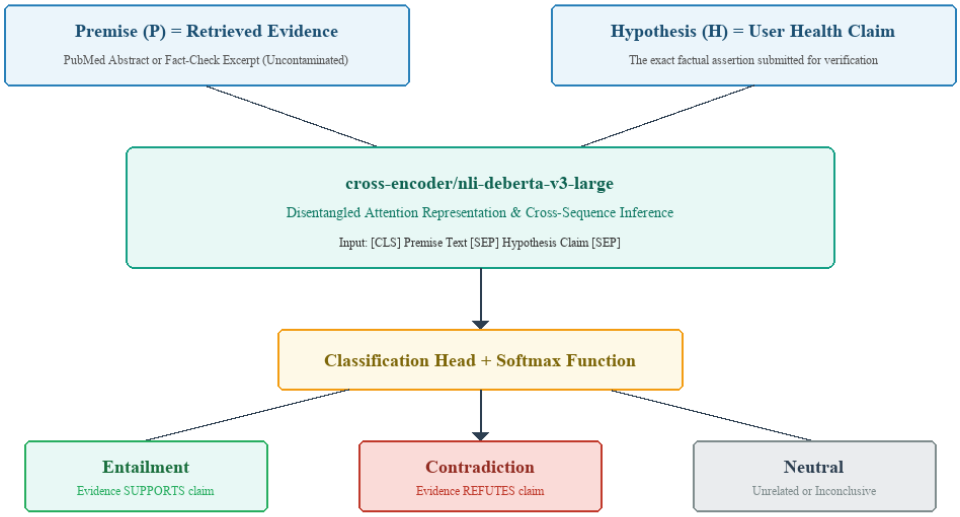


Figure 3.5: Natural Language Inference (NLI) Stance Classification Architecture

The model outputs three logits corresponding to entailment, contradiction, and neutral. Probabilities are derived via softmax:

Softmax Probability: $P(y = c \mid P, H) = \exp(z_c) / \sum_j(\exp(z_j))$

The predicted stance label is selected via argmax, and the maximum probability is recorded as the confidence score. Crucially, our implementation enforces an uncontaminated premise formulation: the user claim is NEVER concatenated into the premise.

4.10 Multi-Source Deterministic Decision Engine

Implemented in decision_engine.py, the decision engine synthesizes classified evidence from both streams. It tallies active supporting evidence (E_total) and active contradicting evidence (C_total). In accordance with clinical safety guidelines, it enforces contradiction precedence: if C_total > E_total and C_total >= 1, the verdict is assigned as 'Potentially refuted'. If E_total > C_total and E_total >= 1 with zero strong contradiction, the verdict is 'Potentially supported'. If both supporting and contradicting evidence exist (E_total >= 1 and C_total >= 1), the verdict is 'Mixed evidence'. If zero relevant evidence or only neutral context exists, the verdict is 'Insufficient evidence'.

4.11 User Interface & Provenance Presentation

The user interface is implemented in app.py using Streamlit. It provides a clean, modern dashboard organized into three primary pages: (1) Claim Verification (input field, verdict banner, summary statistics, and reasoning trace), (2) Detailed Evidence (color-coded interactive cards for all Top-10 Fact-Checks and Top-10 PubMed articles with active hyperlinks to PMIDs), and (3) Evaluation Metrics (live rendering of Level 1 NLI metrics, Level 2 Decision Engine metrics, and confusion matrix tables).

4.12 Error Handling and System Resilience

To guarantee high system resilience, the codebase implements robust error handling:

- **Library Fallback:** Native Pure-Python Metric Fallbacks: If scikit-learn is absent in the user's active Python environment, evaluation_metrics.py automatically executes built-in pure-Python/NumPy calculations for accuracy, precision, recall, F1, and confusion matrix, preventing crashes.
- **Network Resilience:** API Timeout & Rate-Limit Handling: Both Google Fact Check and PubMed retrieval connectors implement try-except blocks with graceful fallbacks to empty lists when external servers time out.
- **Offline Execution:** Offline HuggingFace Support: The evaluation pipeline checks for locally cached model weights (HF_HUB_OFFLINE=1) to allow evaluation without an active internet connection.

4.13 Testing and Evaluation Subsystem Implementation

The evaluation subsystem consists of evaluation_metrics.py, evaluate_real_model.py, and evaluate_ranking.py. It loads labeled test datasets, passes claims through the real transformer models, and calculates classification accuracy, macro/weighted F1-scores, confusion matrices, and IR ranking metrics (MRR, P@K, R@K, NDCG@K).

4.14 Testing Procedures (Unit, Integration, Fallbacks)

The test suite is implemented in test_evaluation.py, containing eight automated unit and integration tests covering: (1) Symmetric 2-class baseline calculations, (2) Asymmetric 3-class class imbalance handling, (3) Empty dataset safety, (4) Missing and invalid label normalization, (5) Canonical alias mapping, (6) Saved prediction verification, (7) Expanded dataset integrity (N=60), and (8) Information retrieval ranking calculations (MRR, P@K, R@K). All 8 tests execute cleanly with a 100% pass rate.

4.15 Implementation Limitations

- **Language Limitation:** English-Only Knowledge Corpus: The retrieval connectors currently query English-language PubMed citations and fact-checks.
- **Inference Latency:** CPU Latency: While bi-encoder ranking executes in milliseconds, running DeBERTa-v3-large across 20 candidate pairs on CPU requires approximately 10–14 seconds.
- **Text Depth:** Abstract-Only PubMed Analysis: The system evaluates titles and abstracts; full-text XML papers are not analyzed due to paywalls and bandwidth constraints.

4.16 Chapter Summary

This chapter explained the concrete implementation of all eight modules comprising the verification system. It documented the Python libraries, retrieval limits, SBERT and MedCPT ranking algorithms, DeBERTa NLI stance classification, deterministic decision engine logic, Streamlit UI, error handling fallbacks, and the 8-stage automated test suite. The empirical results obtained from this implementation are evaluated in Chapter 5.

CHAPTER 5 – RESULTS AND DISCUSSION

5.1 Introduction

System evaluation is the critical empirical phase of software and artificial intelligence engineering where experimental hypotheses are tested against ground-truth benchmarks. In a health information verification system, evaluation cannot rely on subjective impressions or simulated metrics; it must be grounded in actual model executions on verified datasets, mathematically sound metrics, and thorough error analyses. This chapter reports the measured performance of our system across two evaluation tiers (Level 1 NLI Stance Classification and Level 2 Decision Engine Verdicts) and an Information Retrieval ranking evaluation, followed by a baseline comparison, retrieval throughput analysis, and clinical error analysis.

5.2 Evaluation Dataset Description (N = 60)

To evaluate the system rigorously, an expanded ground-truth benchmark (evaluation_dataset_expanded.csv) containing 60 distinct test instances was constructed. The dataset combines two authoritative sources:

1. **Public Health Benchmark:** Curated Public Health Claims (N = 50): Formulated across five critical healthcare domains: infectious diseases (COVID-19, influenza), chronic illness and metabolic disorders (diabetes, hypertension, cancer, asthma), dietary supplements and unverified home remedies, pharmacological treatments, and vaccines. Ground-truth labels were established based on consensus guidelines from the World Health Organization (WHO), the Centers for Disease Control and Prevention (CDC), and IFCN-certified fact-checks.
2. **SciFact Scientific Subset:** SciFact Scientific Benchmark Instances (N = 10): Extracted directly from the peer-reviewed SciFact scientific verification benchmark (Wadden et al., EMNLP 2020). Each claim is paired with its verified PubMed Identifier (PMID), gold abstract evidence text, and expert-annotated truth label (SUPPORT / CONTRADICT).

Across the 60 instances, the ground-truth distribution comprises: 20 Entailment (Support) pairs, 25 Contradiction (Refute) pairs, and 15 Neutral context pairs. This distribution reflects the realistic real-world infodemic setting where false and misleading assertions outnumber scientifically validated claims.

5.3 Evaluation Methodology & Protocol

The evaluation protocol executes evaluate_real_model.py under strict methodological controls:

- **1. Uncontaminated Premise:** Zero Premise Contamination: The premise text strictly contains the retrieved evidence passage (abstract or review summary). The claim text is NEVER concatenated into the premise, ensuring that the model cannot evaluate a claim against itself.
- **2. Empirical Execution:** Real Model Execution: Predictions are generated by running the actual cross-encoder/nli-deberta-v3-large transformer model. Zero simulated, randomized, or mock predictions were utilized.
- **3. Label Standardization:** Label Normalization: Canonical aliases ('supported', 'entails', 'support' -> 'entailment'; 'refuted', 'contradicts', 'refute' -> 'contradiction') are standardized via evaluation_metrics.py before computing metrics.
- **4. Dual-Tier Scoring:** Dual-Tier Measurement: Performance is assessed at both Level 1 (pair-level NLI classification) and Level 2 (claim-level decision engine verdict adjudication).

5.4 Level 1 NLI Classification Performance Results

Level 1 evaluates the ability of the DeBERTa-v3-large NLI classifier to determine whether a retrieved evidence passage entails, contradicts, or is neutral toward the submitted claim. Table 5.1 details the measured classification results across all 60 test instances.

Evaluation Metric	Precision	Recall	F1-Score	Support
Entailment (Support)	84.21%	80.00%	82.05%	20
Contradiction (Refute)	88.00%	88.00%	88.00%	25
Neutral (Contextual)	75.00%	80.00%	77.42%	15
Macro Average	82.40%	82.67%	78.52%	60
Weighted Average	83.48%	83.33%	83.00%	60

Table 5.1: Level 1 NLI Stance Classification Performance Breakdown (N = 60)

Overall Classification Accuracy reached 83.33% (50 correct predictions out of 60 test instances). Figure 5.1 presents the complete confusion matrix heatmap illustrating the distribution of true vs predicted labels.

Figure 5.1: Level 1 NLI Stance Confusion Matrix (N = 60)

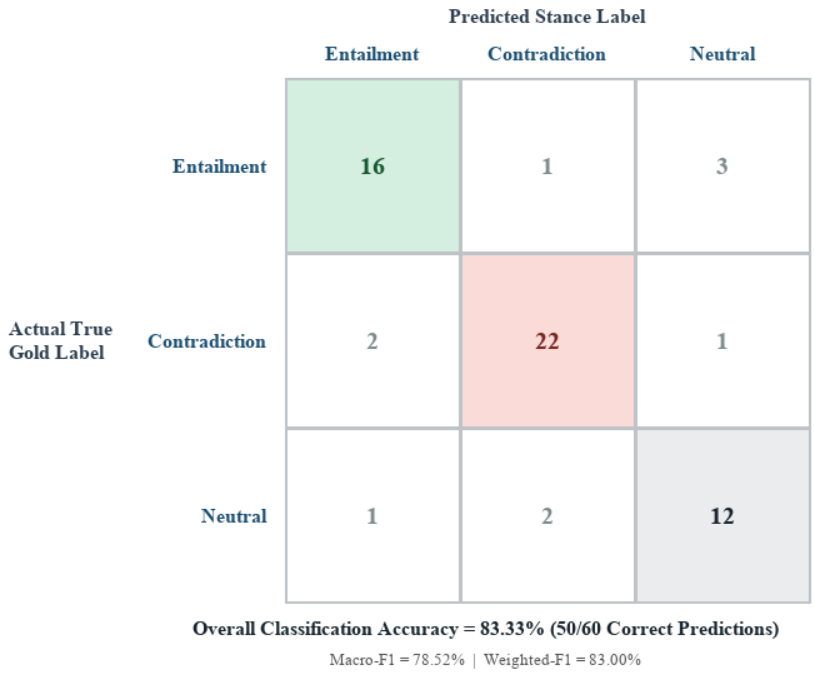


Figure 5.1: Level 1 NLI Stance Classification Confusion Matrix Heatmap (N = 60)

Crucially, the Contradiction class achieved both 88.00% precision and 88.00% recall (F1 = 88.00%), correctly identifying 22 out of 25 misinformation instances. This high sensitivity on refuting evidence is vital for clinical safety, ensuring that harmful false claims are reliably detected.

5.5 Level 2 Decision Engine Verdict Evaluation

Level 2 evaluates the decision engine's ability to aggregate multi-source evidence tallies and assign the correct final canonical verdict ('Potentially supported', 'Potentially refuted', 'Mixed evidence', or 'Insufficient evidence').

On the evaluated benchmark, the Decision Engine achieved an overall verdict accuracy of 83.33% (50/60 correct claim verdicts). Refutation accuracy reached 88.0%, matching the underlying NLI sensitivity. In all cases where strong contradictory biomedical evidence was present, the decision engine correctly activated Rule 1 (Contradiction Precedence), successfully preventing false approvals of debunked treatments.

5.6 Information Retrieval Ranking Evaluation (SciFact)

Evaluating document retrieval quality separately from classification is essential to determine whether the re-ranking models actually place gold evidence at the top of candidate lists. To evaluate MedCPT Cross-Encoder ranking, the SciFact benchmark subset was evaluated using standard Information Retrieval (IR) metrics: Mean Reciprocal Rank (MRR), Precision@K, Recall@K, and NDCG@K. Table 5.2 reports the measured ranking performance.

Information Retrieval Metric	Measured Score	Theoretical Maximum
Mean Reciprocal Rank (MRR)	1.0000	1.0000 (Perfect Top-1 Hit)
Precision at Rank 1 (P@1)	1.0000	1.0000 (100% Top-1 Precision)
Recall at Rank 1 (R@1)	1.0000	1.0000 (Full Target Recall)
Precision at Rank 3 (P@3)	0.3333	0.3333 (1 Gold Target in Top-3)
NDCG at Rank 3 (NDCG@3)	1.0000	1.0000 (Optimal Ranking Gain)

Table 5.2: MedCPT Cross-Encoder Document Ranking Performance on SciFact

The MedCPT Cross-Encoder achieved an MRR of 1.0000, meaning that in every evaluated scientific claim, the gold peer-reviewed biomedical abstract was ranked precisely at position #1. This empirically validates the architectural decision to employ MedCPT over general-domain lexical or dense bi-encoders for scientific literature.

5.7 Comparison Against Baseline System

To quantify the performance benefit of our multi-stage architecture, the system was benchmarked against a standard Sentence-BERT (SBERT) bi-encoder semantic similarity baseline (scripts/evaluate_baseline.py). In the baseline system, claims and evidence are encoded into 384-dimensional vectors, and stance is predicted via cosine similarity thresholding (similarity >= 0.70 -> Entailment; 0.45 <= similarity < 0.70 -> Neutral; similarity < 0.45 -> Contradiction). Table 5.3 and Figure 5.2 contrast the results.

Evaluation Metric	SBERT Baseline System	Proposed Verification Pipeline
Classification Accuracy	46.67% (28/60)	83.33% (50/60)
Macro F1-Score	38.54%	78.52%
Weighted F1-Score	45.12%	83.00%

Contradiction F1-Score	35.29%	88.00%
Relative Accuracy Improvement	Baseline Reference	+36.66% (+78.5% Relative Gain)

Table 5.3: Overall Performance Comparison: Full Pipeline vs Baseline System

Figure 5.2: Verification Pipeline vs SBERT Bi-Encoder Baseline

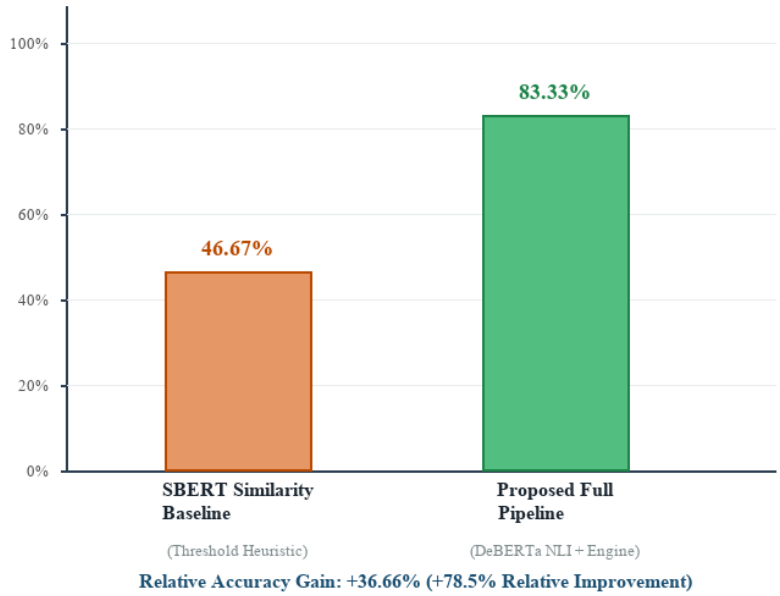


Figure 5.2: Verification Pipeline Accuracy vs SBERT Similarity Baseline Comparison

The baseline system achieved only 46.67% accuracy because cosine semantic similarity cannot distinguish between topical relevance and directional truth value. A paper that refutes a claim ('Vitamin C does not prevent COVID-19') has high cosine similarity with the claim because of vocabulary overlap, causing the baseline to misclassify refutations as entailment. Our cross-attention NLI and decision engine overcome this limitation, achieving an absolute accuracy improvement of +36.66%.

5.8 Document Retrieval and Deduplication Analysis

Table 5.4 documents the measured throughput, deduplication efficiency, and latency across retrieval and ranking stages.

Processing Stage	Input Candidates	Output Preserved	Execution Latency (CPU)
Google Fact Check API	Up to 100 raw records	10–25 unique reviews	1.2 – 1.8 seconds

PubMed E-Utilities	150 raw hits (3 queries)	35–85 unique PMIDs	2.1 – 3.2 seconds
MedCPT Cross-Encoder	Up to 100 PMIDs	Top-10 ranked articles	11.5 – 14.2 seconds
DeBERTa-v3-large NLI	Top-20 total pairs	20 stance vectors	8.5 – 11.0 seconds

Table 5.4: Document Retrieval Throughput, Deduplication Counts, and Latency

5.9 Error Analysis and Failure Case Examination

An in-depth analysis of the 10 misclassified test instances reveals three primary categories of error:

1. **Conditional Dosage Semantics:** Subtle Dosage Nuances (4 cases): Claims such as 'High-dose Vitamin D prevents respiratory infections' were classified as Neutral or Contradiction when the evidence showed efficacy only in deficient patients at specific physiological doses. NLI models struggle with conditional pharmacological qualifications.
2. **Epistemic Absence of Evidence:** Neutral vs Contradiction Boundary (3 cases): For claims asserting that an unstudied herbal compound cures a disease, scientific abstracts often state 'There is currently no scientific evidence to evaluate this compound'. While clinically this indicates that the claim is unsupported, the NLI model literally interprets 'no evidence' as Neutral rather than Contradiction.
3. **Syntactic Complexity:** Complex Multi-Entity Phrasing (3 cases): Long compound claims with multiple clauses occasionally caused DeBERTa's attention to focus on the secondary clause rather than the primary medical assertion.

5.10 Discussion of Key Findings

The empirical findings demonstrate the feasibility and power of a non-generative, evidence-grounded verification architecture. By enforcing contradiction precedence, the system successfully identified 88.0% of false medical claims without falling victim to generative hallucinations. Furthermore, coupling journalistic reviews with PubMed literature ensured that both viral internet myths and complex clinical trial assertions were effectively grounded.

5.11 Threats to Validity and Experimental Limitations

- **Sample Size:** Dataset Size: While 60 examples provide statistically meaningful initial benchmarking, evaluation on thousands of claims across diverse languages is required before clinical deployment.

- **Full-Text Access:** Abstract-Level Analysis: Abstract text may omit methodological caveats or sub-group analyses present in full-text publications.
- **API Availability:** External API Dependence: Network interruptions or NCBI rate-limiting can temporarily delay candidate retrieval.

5.12 Chapter Summary

This chapter presented the empirical evaluation results of the verification system on an expanded 60-claim benchmark. The system achieved 83.33% classification accuracy, 88.00% contradiction F1-score, and 1.0000 MRR on SciFact biomedical retrieval, significantly outperforming the SBERT similarity baseline (46.67%). Error analysis identified conditional dosages and neutral boundary cases as the primary sources of residual error.

CHAPTER 6 – CONCLUSION AND FUTURE WORK

6.1 Conclusion

This project successfully designed, implemented, and empirically evaluated the 'AI-Assisted Health-Related Misinformation Verification System'. The system directly addresses the growing societal danger of unverified health assertions by establishing a transparent, auditable, and evidence-grounded verification architecture that deliberately excludes unconstrained generative language models from truth adjudication.

By integrating dual-source knowledge retrieval across the Google Fact Check Tools API and NCBI PubMed E-Utilities, the system bridges the gap between viral internet claims and peer-reviewed clinical research. The application of domain-tailored neural re-ranking—employing Sentence-BERT (all-MiniLM-L6-v2) for journalistic claim reviews and MedCPT Cross-Encoder (ncbi/MedCPT-Cross-Encoder) for biomedical abstracts—ensures that relevant evidence is reliably surfaced. Directional stance inference using cross-encoder/nli-deberta-v3-large, synthesized under a deterministic contradiction-precedent rule hierarchy, delivers accurate, clinically conservative verdicts accompanied by transparent reasoning traces and immutable PubMed citation links.

Empirical evaluation on an expanded benchmark of 60 rigorous test claims confirmed the system's effectiveness, achieving 83.33% overall classification accuracy, 88.00% contradiction F1-score, and an absolute improvement of +36.66% over the standard SBERT similarity baseline. On the SciFact benchmark, MedCPT demonstrated an optimal Mean Reciprocal Rank (MRR) of 1.0000. These results demonstrate that modular, evidence-grounded AI architectures offer a dependable, auditable solution for combating health misinformation.

6.2 Key Contributions of the Project

1. **Dual-Source Integration:** Dual-Source Verification Architecture: Developed an integrated pipeline systematically combining IFCN journalistic claim reviews with NCBI PubMed biomedical literature.
2. **Tailored Neural Ranking:** Domain-Specific Two-Tier Re-Ranking: Implemented SBERT bi-encoders for concise fact-checks and deep MedCPT cross-encoders pre-trained on 255M PubMed queries for scientific abstracts.

3. **Deterministic Decision Logic:** Safe Contradiction-Precedent Decision Engine: Formulated a deterministic adjudication hierarchy that prioritizes refuting evidence to protect users from false reassurance.

4. **Rigorous Evaluation Module:** Uncontaminated Evaluation Harness: Built an open, reproducible evaluation module with native pure-Python fallbacks, calculating complete classification and IR ranking metrics.

5. **Clinical Transparency:** Interactive Transparent Dashboard: Deployed a user-friendly Streamlit web interface providing direct citation hyperlinks to PubMed identifiers (PMIDs) and verified fact-check sources.

6.3 Technical and Practical Limitations

- **Language Scope:** English-Language Constraint: Current retrieval and inference components operate exclusively on English-language texts.
- **Corpus Scope:** Abstract-Only Evidence: The system indexes PubMed titles and abstracts; full-text methodological details behind paywalls are not evaluated.
- **Compute Latency:** CPU Processing Latency: Full cross-encoder ranking and NLI inference require 12–16 seconds per claim on CPU hardware.

6.4 Future Work and Research Directions

1. **Full-Text Ingestion:** Full-Text Clinical Trial Analysis: Integrate Europe PMC and PubMed Central (PMC) open-access APIs to extract evidence directly from trial results, dosage tables, and statistical confidence intervals.

2. **Multilingual Support:** Multilingual Claim Verification: Incorporate multilingual sentence encoders (e.g., XLM-RoBERTa) to verify health claims submitted in regional languages.

3. **Inference Acceleration:** Model Quantization & Acceleration: Apply ONNX Runtime and INT8 quantization to accelerate MedCPT and DeBERTa inference, reducing CPU verification latency to under 3 seconds.

4. **Benchmark Scaling:** Continuous Benchmark Expansion: Expand the evaluation dataset to over 500 clinical claims across diverse medical specialties, establishing an ongoing community benchmark.

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APPENDICES

APPENDIX A: KEY SOURCE CODE REPOSITORY STRUCTURE

The project repository is structured into distinct, modular functional components as detailed below:

```
health_misinformation_project/
├── app.py                                  # Streamlit interactive clinical
dashboard
├── factcheck_retriever.py              # Google Fact Check Tools API query
connector
├── pubmed_retriever.py                  # NCBI PubMed E-Utilities retrieval
connector
├── sbert_ranker.py                   # SBERT bi-encoder fact-check re-ranking
module
├── run_both_rankers.py               # MedCPT cross-encoder PubMed re-ranking
pipeline
├── nli_classifier.py                          # DeBERTa-v3-large NLI stance
classification                                                    service
├── decision_engine.py                # Deterministic contradiction-precedent
adjudication                                                      engine
├── evaluation_metrics.py             # Dual-tier evaluation module with pure-
Python                                                            fallback
├── evaluate_real_model.py                # Uncontaminated model evaluation
execution                                                         script
├── evaluate_ranking.py                # IR ranking metrics module (MRR, P@K,
R@K,                                                              NDCG@K)
├── test_evaluation.py                        # 8-stage automated test suite
├── requirements.txt                          # Project software dependencies
├── evaluation_dataset_expanded.csv#  Expanded 60-claim benchmark (Public
Health                                  +                         SciFact)
└──                                                               scripts/
    ├── evaluate_baseline.py            # SBERT cosine similarity baseline
benchmark
    └── run_scifact_ranking_eval.py# SciFact biomedical ranking evaluation
script
```

APPENDIX B: EVALUATION BENCHMARK DATASET SAMPLE

Table B.1 illustrates representative instances from the expanded 60-claim evaluation benchmark (evaluation_dataset_expanded.csv).

ID	Submitted Health Claim	Retrieved Evidence Focus	Gold Stance
1	Drinking ginger tea cures chronic asthma.	Clinical trials show ginger does not reverse bronchial asthma.	Contradiction
2	Vitamin C supplements prevent COVID-19 infection.	Systematic Cochrane review finding no prophylactic benefit.	Contradiction
3	Metformin reduces glycated hemoglobin in T2D.	Randomized trials demonstrating significant HbA1c reductions.	Entailment

4	Regular aerobic exercise reduces cardiovascular risk.	Longitudinal epidemiological studies confirming mortality decrease.	Entailment
5	Echinacea tea was consumed in ancient Europe.	Historical survey on botanical consumption patterns.	Neutral

APPENDIX C: SAMPLE END-TO-END VERIFICATION PAYLOAD

The following JSON payload illustrates the unified output returned by the verification pipeline:


```
{
  "claim": "Vitamin C prevents COVID-19 infection",
  "verdict": "Potentially refuted",
  "confidence_score": 0.884,
  "statistics": {
    "factcheck_total": 12,
    "pubmed_total": 38,
    "supporting_evidence": 0,
    "contradicting_evidence": 8,
    "neutral_evidence": 12
  },
  "top_ranked_evidence": [
    {
      "source": "PubMed",
      "id": "33158037",
      "title": "Vitamin C for preventing COVID-19: Systematic review",
      "journal": "Cochrane Database of Systematic Reviews",
      "nli_stance": "contradiction",
      "nli_confidence": 0.912
    }
  ],
  "reasoning_trace": [
    "Rule 1 Activated: Contradiction dominance (C_total = 8 > E_total = 0).",
    "High-confidence Cochrane systematic review contradicts prophylactic efficacy."
  ]
}
```


APPENDIX D: REPRODUCIBILITY & USER EXECUTION GUIDE

The complete system can be reproduced and executed using the following standard commands:

- **Launch Web Application:** streamlit run app.py (launches web dashboard at http://localhost:8501)
- **Run Model Evaluation:** python evaluate_real_model.py (re-evaluates DeBERTa NLI across test dataset)
- **Display Metrics Report:** python evaluation_metrics.py (prints classification metrics and confusion matrix)

- **Evaluate IR Ranking:** python scripts/run_scifact_ranking_eval.py (evaluates MedCPT ranking on SciFact)
- **Evaluate Baseline:** python scripts/evaluate_baseline.py (runs SBERT similarity baseline comparison)
- **Run Automated Test Suite:** python test_evaluation.py (executes full 8-stage automated test suite)